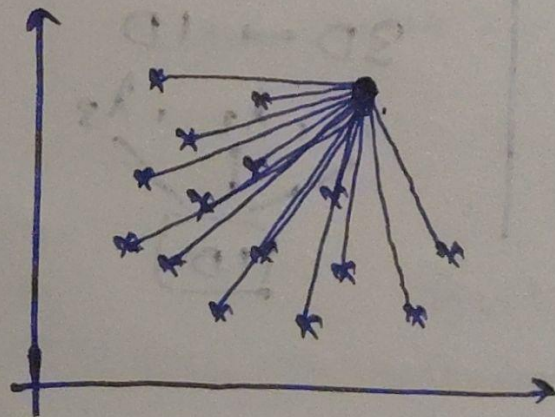
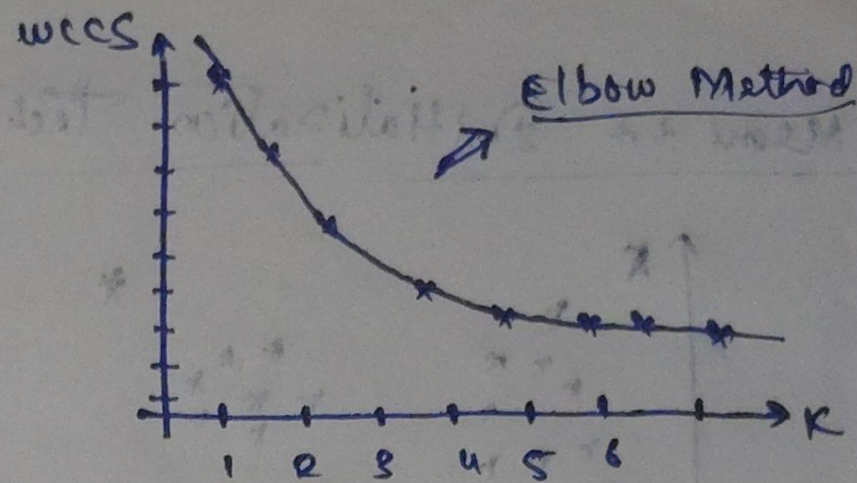
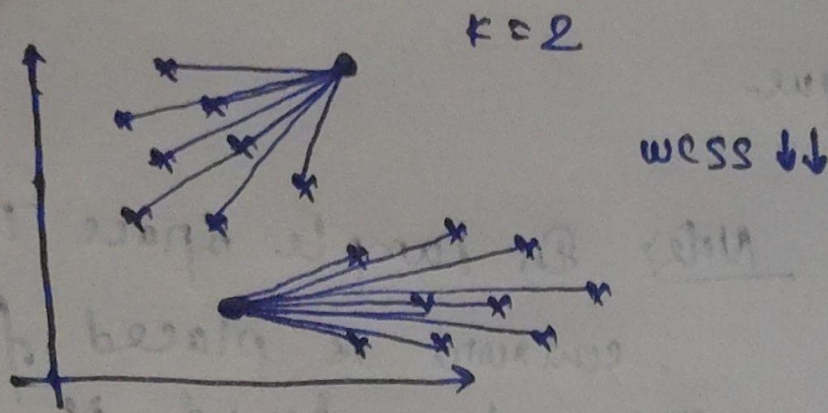


WCSS = within cluster Sum of squares.

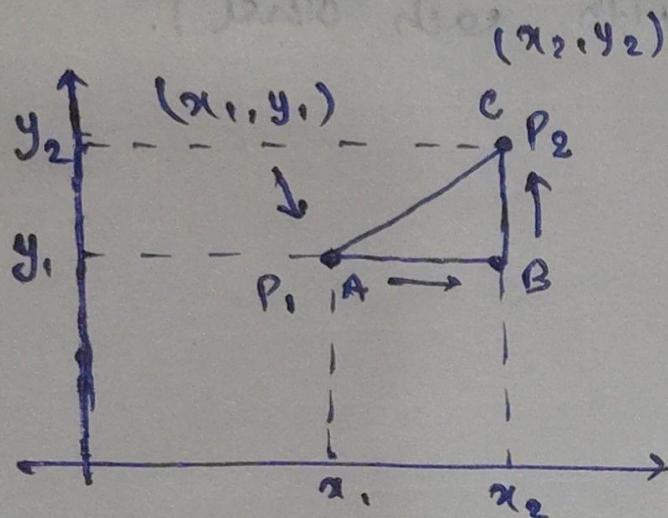
Initialize $k=1$ to 20



$$WCSS = \sum_{i=1}^n (\text{distance between point } i \text{ to nearest centroid})^2$$



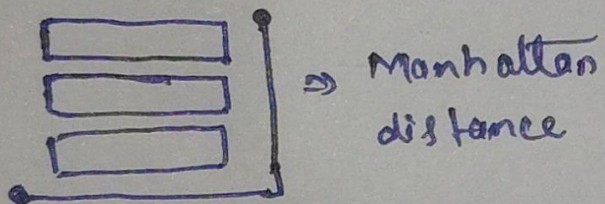
* Euclidean Distance:



$$\text{Euclidean dist.} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Manhattan dist.} = |x_2 - x_1| + |y_2 - y_1|$$

IRON MAN → U.S.



Air Traffic

